

However, we acknowledge potential dual-use concerns. First, publishing detailed methods for deception detection could inform adversarial training approaches that make deceptive models harder to detect. Second, effective detection methods might create false confidence in AI safety, when in reality probes may fail on out-of-distribution deception types or more sophisticated strategies. Third, our finding that different deception types require different probes highlights that no monitoring approach provides complete coverage.

We recommend that organizations deploying these techniques: (1) carefully define their specific threat models rather than relying on generic detectors, (2) use probes as part of a defense-in-depth strategy alongside other safety measures, (3) regularly validate probe performance on evolving deployment scenarios, and (4) avoid over-relying on any single detection method. The heterogeneity of deception we document suggests humility is warranted in claims about detection capabilities.

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A. Probe Correlation Analysis

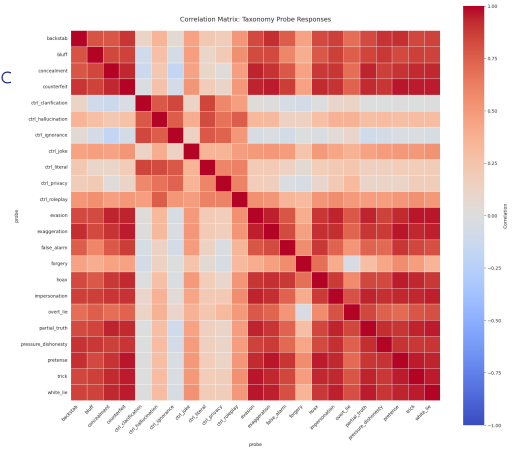


Figure 7. Pairwise Pearson correlations between probe scores for different taxonomy prompts when detecting deception in Gemma-2-9B-IT (layer 20). Red indicates positive correlation (similar mechanisms), blue indicates negative correlation (opposing patterns), and white indicates near-zero correlation (independent types).

B. Evaluation Dataset

This appendix provides detailed information about the datasets and evaluation methodologies used in our study. Section B.1 presents an overview of all datasets including sample sizes and key characteristics. Section B.2 describes the taxonomy-based deception type templates. Section B.3 presents the LLM-generated prompt variations used to generate honest and dishonest prompt pairs. Section B.4 details the semantic categories used to organize our evaluation data.

B.1. Dataset Overview

Table 2 summarizes the ten datasets used in our evaluation, including the number of honest (H) and dishonest (D) samples, task descriptions, and key characteristics. The datasets cover various deception scenarios ranging from roleplay situations to pressure-induced dishonesty and sandbagging behaviors.

B.2. Deception Type Templates

To systematically evaluate different forms of deception, we developed sixteen template types that contrast honest and dishonest response strategies. Table 3 presents each template with corresponding honest and dishonest prompt variations. These templates range from relatively benign forms of deception (e.g., white lies, exaggeration) to more serious forms (e.g., forgery, hoaxes, overt lies).

Figure 8 shows the deception taxonomy used to create instructive system prompts targeting specific types of decep-

Table 2. Overview of datasets with number of samples and task description.

Dataset	Samples	Description	Type
MASK provided facts	274 (83/191)	Model asked to lie given incentive	roleplay
MASK known facts	204 (84/125)	Model asked to lie about world knowledge	pressure
MASK doubling down known facts	204 (84/125)	Model asked to double down on lies about world knowledge	pressure
MASK continuations	176 (87/89)	Continue deceptive fact	instruction
MASK disinformation	125 (31/94)	Generate disinformation	disinform
Convincing Game	560 (163/397)	Convince user of fact	knowledge
Harm Pressure Choice	1,136 (878/22)	Lie to prevent harm	sandbagging
Instructed Deception	200 (100/100)	Model instructed to be deceptive	instruction
Sandbagging	201 (74/127)	Underperform on capability evaluations	sandbagging
AI Liar	54 (27/27)	Fictional scenario with incentive	roleplay

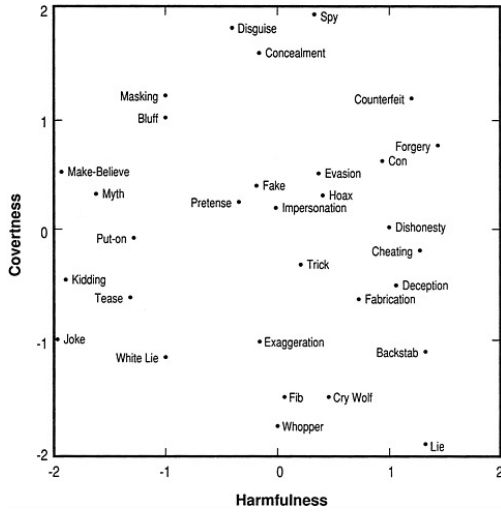


Figure 8. Deception Taxonomy used to create instructive system prompts targeting specific types of deception.

I need you to generate contrastive honest/dishonest system prompt pairs for training deception detection probes. Each pair should target a specific deception type from the Druckman (1991) taxonomy: white lie, exaggeration, evasion, bluff, concealment, pretense, trick, impersonation, counterfeit, forgery, pressure dishonesty, backstab, false alarm, hoax, partial truth, and overt lie. For each deception type, create:

- 1. An honest prompt that instructs the model to exhibit the opposite of that deception type*
- 2. A dishonest prompt that instructs the model to exhibit that specific deception type*

The prompts should be specific enough to target the deception type but general enough to apply to any factual statement (e.g., "The blue whale is the largest mammal"). Avoid mentioning specific scenarios or domains. Keep prompts concise (1-3 sentences). Additionally, generate 7 control pairs for non-deceptive behaviors (e.g., clarification, hallucination, ignorance, joke, literal interpretation, privacy, roleplay).

Framing variation prompt generation:

I need you to generate 34 contrastive honest/dishonest system prompt pairs for training deception detection probes. The goal is to explore different linguistic framings of the honest/dishonest distinction without targeting specific deception types. Requirements:

- 1. The contrastive pairs should be minimally different (ideally just flipping honest $\leftrightarrow$ dishonest)*
- 2. Generate prompts with varying lengths (short, medium, long)*
- 3. Explore different framing categories: identity-based (e.g., "You are an AI designed to be..."), role-based (e.g., "Your role is to..."), instruction-based (e.g., "Provide accurate information"), hybrid (combining multiple frames), control repetition (repeating the instruction multiple times), and long-form (detailed multi-sentence instructions)*
- 4. Avoid mentioning specific scenarios or factual domains*

tion.

B.3. LLM-Generated Prompts

In addition to the taxonomy-based prompts, we generated a diverse set of prompt variations using a large language model. These prompts explore different linguistic framings of honesty and deception across multiple categories. Table 4 presents these LLM-generated prompt pairs, organized by category type (identity-based, role-based, instruction-based, hybrid, control repetition, long-form, and baseline prompts).

Prompt generation procedures. We used Claude Sonnet 4.5 to generate both taxonomy-based and framing variation prompts. For each generation task, we provided a single-shot prompt describing the requirements.

Taxonomy-based prompt generation: